Machine Learning Lab
Module 01

Starting Machine Learning with Structured Data

Lecture Note • Structured Data • Case Study

Based on the paper: "Early Prediction of Diabetes Using an Ensemble of Machine Learning Models" (2022).

Purpose: To understand, step by step, how a structured-data machine-learning pipeline is designed, compared, optimized and validated.

1. Learning Objectives

After studying this module, students should be able to explain a complete machine-learning workflow for structured data; identify the role of data preparation, feature selection, model training and evaluation; describe why different preprocessing strategies should be compared experimentally; and explain how individual classifiers can be combined into a weighted ensemble.

2. Paper at a Glance

The paper, Early Prediction of Diabetes Using an Ensemble of Machine Learning Models, develops a machine-learning pipeline for early diabetes prediction using population-level data from Bangladesh. Rather than relying on a single classifier, the study investigates preprocessing, feature selection, hyperparameter optimization, several individual classifiers and weighted ensembles. The final system is evaluated using cross-validation and statistical testing.

The methodology is particularly useful as a laboratory case study because each major stage can be reproduced as an independent machine-learning experiment.

3. Methodology: The Complete Workflow

The study follows a progressive experimental workflow: BDHS data → construction of DDC datasets → diabetes-label definition → missing-value experiments → feature-selection experiments → hyperparameter optimization → individual classifiers → weighted ensemble → cross-validation → statistical validation.

The important point is that the authors did not select every component at the beginning. They compared alternatives at several stages and carried the best-performing choice into the next stage.

4. Step 1 — Dataset Construction

The authors constructed Diabetes Diseases Classification (DDC) datasets from the Bangladesh Demographic and Health Survey (BDHS). Two survey sources were used: BDHS 2011 and BDHS 2017–2018.

DDC-2011 contains 4,751 diabetes and 2,814 non-diabetes samples. DDC-2017 contains 3,492 diabetes and 4,073 non-diabetes samples. The resulting task is binary classification.

The BDHS data contain demographic, socioeconomic, lifestyle and health information collected from respondents. The authors use these population-level variables to build a diabetes-prediction dataset.

5. Step 2 — Defining the Target

Diabetes status is used as the target variable. The paper reports that fasting blood glucose of at least 7.0 mmol/L is treated as diabetes. Values from 6.0 to 6.9 mmol/L and values below 6.0 mmol/L are placed in the non-diabetes category.

Thus, the learning problem can be represented as $Y \in \{0, 1\}$, where the two classes represent diabetes and non-diabetes.

6. Step 3 — Selecting the Input Features

The initial dataset contains 13 features covering geographic, demographic, socioeconomic, lifestyle and health-related information. The variables include division, urban/rural residence, wealth index, gender of household head, age, education, occupation, eating habit, caffeinated-drink consumption, smoking, systolic measurement, diastolic measurement and BMI.

The purpose is to predict diabetes from information that can be available at the population or individual level rather than using the glucose measurement itself as an input feature.

7. Step 4 — Missing-Value Handling

The authors recognize that missing observations are common in real-world medical datasets. Three approaches are compared: case deletion, MEDimpute and KNNimpute.

Case deletion removes observations containing missing values. MEDimpute replaces missing values using the median. KNNimpute estimates missing values from similar observations using nearest-neighbor information.

The authors compare the alternatives using downstream classifier performance rather than assuming that one imputation method is universally best.

8. Step 5 — Imputation Ablation Study

The three missing-value strategies are tested on the DDC datasets using several classifiers and AUC as a principal comparison measure. MEDimpute generally gives stronger results and is selected for the subsequent experiments.

This is an important experimental principle: a preprocessing method is selected because it performs well in the intended learning pipeline.

9. Step 6 — Feature Selection

After missing-value handling, the authors investigate whether all 13 variables are necessary. Four feature-selection approaches are considered: Random Forest (RF) feature importance, Information Gain (IG), XGBoost (XGB) feature importance and LightGBM (LGB) feature importance.

The reason for feature selection is to reduce unnecessary or redundant variables and to determine whether a smaller feature set can maintain or improve predictive performance.

10. Step 7 — RF-Based Feature Selection

Random Forest is used to rank variables according to feature importance. The five features selected in the main pipeline are BMI (F13), age (F5), systolic measurement (F11), diastolic measurement (F12) and occupation (F7).

The authors also vary the number of retained features and compare the resulting AUC values. RF-based selection is chosen because it provides strong performance with a relatively small feature subset.

11. Step 8 — Hyperparameter Optimization

Machine-learning models can be highly sensitive to hyperparameter choices. The authors therefore use Grid Search Optimization to find suitable parameter combinations for the classifiers.

Grid search evaluates predefined combinations of hyperparameters and selects the combination that gives the best performance according to the experimental evaluation.

12. Step 9 — Individual Classifiers

Six classifiers are studied: Gaussian Naive Bayes (GNB), Bernoulli Naive Bayes (BNB), Random Forest (RF), Decision Tree (DT), XGBoost (XGB) and LightGBM (LGB).

The models represent different learning approaches, including Bayesian, tree-based and boosting methods. Comparing different model families helps determine whether one model consistently outperforms the others and provides candidates for ensemble learning.

13. Step 10 — Performance Evaluation

The paper evaluates models using sensitivity (Sn), specificity (Sp), accuracy (Acc) and area under the ROC curve (AUC).

Sensitivity measures the proportion of actual diabetes cases detected. Specificity measures the proportion of non-diabetes cases correctly identified. Accuracy measures overall correct classification, while AUC summarizes discrimination across classification thresholds.

The multiple metrics are important because a model can show high sensitivity while having very poor specificity. The paper reports examples where Random Forest reaches 100% sensitivity but 0% specificity, making it unreliable despite the apparently strong sensitivity.

14. Step 11 — Weighted Ensemble Learning

The central methodological contribution is a weighted ensemble. Instead of treating all classifiers equally, the authors use each model's AUC as a weight.

If model j produces a class probability Pij and has weight Wj, the ensemble combines the model probabilities according to their weights. The class with the highest resulting ensemble probability is selected.

The motivation is straightforward: a classifier with stronger discrimination should have greater influence on the final prediction than a weaker classifier.

15. Step 12 — Ensemble Ablation Study

The authors systematically test different combinations rather than simply combining all six classifiers.

Pairwise combinations include GNB + BNB, RF + DT and LGB + XGB. Four-model combinations include GNB + BNB + RF + DT, RF + DT + LGB + XGB and GNB + BNB + XGB + LGB. A six-model combination containing all classifiers is also tested.

The experiments show that adding more models does not automatically improve performance. In some cases, a four-model ensemble performs better than the six-model ensemble.

16. Step 13 — Final Ensemble

For the main proposed system, the weighted combination of Decision Tree, Random Forest, XGBoost and LightGBM is identified as the strongest ensemble configuration in the reported results.

The abstract reports an accuracy of 0.735 and an AUC of 0.832 for the proposed ensemble with the preprocessing strategy used in the study.

17. Step 14 — Stratified 5-Fold Cross-Validation

The authors use stratified 5-fold cross-validation to evaluate the robustness of the classifiers. The dataset is divided into five folds while preserving the relative class proportions. Four folds are used for training and one for validation, and the process is repeated five times.

Cross-validation reduces dependence on a single train-test split and provides a more reliable estimate of model performance.

18. Step 15 — ANOVA and Statistical Validation

The authors apply ANOVA to determine whether the observed differences among model performances are statistically meaningful. They use $\alpha = 0.05$ as the significance threshold.

The reported ANOVA p-value is 3.52×10^{-3}, which is below 0.05. The null hypothesis that all model means are identical is therefore rejected. The paper also uses post-hoc testing to identify which model or ensemble is superior.

19. Step 16 — Year-Wise Generalization Experiment

The authors perform an additional experiment to examine whether selected features remain useful across survey years. In Case 1, features selected using DDC-2017 are applied to DDC-2011. In Case 2, features selected using DDC-2011 are applied to DDC-2017. In Case 3, the datasets are merged and feature selection is performed on the combined data.

The repeated appearance of BMI, age, systolic and diastolic measurements among the leading variables provides evidence that these predictors are relatively stable within the studied datasets.

20. Complete Pipeline

1. Obtain BDHS 2011 and BDHS 2017–2018 data.
2. Construct the DDC-2011 and DDC-2017 datasets.
3. Define the binary diabetes target.
4. Prepare the 13 candidate input features.
5. Compare case deletion, MEDimpute and KNNimpute.
6. Select MEDimpute for subsequent experiments.
7. Compare RF, IG, XGB and LGB feature-selection strategies.

8. Select the RF-based feature set.
9. Optimize classifier hyperparameters using grid search.
10. Train GNB, BNB, RF, DT, XGB and LGB.
11. Compare individual model performance.
12. Construct AUC-weighted ensemble combinations.
13. Select the strongest ensemble configuration.
14. Evaluate using stratified 5-fold cross-validation.
15. Apply ANOVA and post-hoc statistical analysis.
16. Test feature generalization across survey years.

21. Key Lessons for a Machine Learning Laboratory

This paper is a useful example of an end-to-end structured-data workflow. The main lesson is not simply that an ensemble can outperform an individual model. The stronger lesson is the experimental process used to reach that model.

First, preprocessing choices should be tested. Second, feature-selection choices should be tested. Third, hyperparameters should be tuned rather than left at arbitrary defaults. Fourth, individual models should be compared before constructing an ensemble. Fifth, an ensemble should be justified through ablation experiments. Finally, the reported improvement should be examined using cross-validation and statistical testing.

The study also demonstrates why accuracy alone is insufficient. Sensitivity, specificity and AUC can reveal failure modes that accuracy may hide.

22. Methodological Note

The paper mentions outlier rejection among its preprocessing contributions. However, the detailed methodology available in the paper explicitly describes the missing-value strategies, feature-selection procedures and subsequent modeling stages more clearly than it describes a standalone outlier-detection and removal procedure. Therefore, an implementation should not invent a particular outlier algorithm unless the authors' source or supplementary material specifies it.

23. Discussion Questions

17. Why is median imputation preferred to deleting every record with a missing value?
18. Why might RF feature selection produce a different feature ranking from Information Gain?
19. Why can 100% sensitivity still represent a poor classifier?
20. Why is AUC suitable for assigning ensemble weights in this study?
21. Why does adding all six classifiers not necessarily improve the ensemble?
22. What does 5-fold cross-validation tell us that a single train-test split cannot?
23. What does the ANOVA p-value contribute beyond reporting the AUC values?
24. What evidence would be needed to reproduce the paper exactly?